04 / PSEUDOCODE GUIDE

From geometric transformation to algorithm

EX29 · Designing before coding

Purpose
Before programming, identify which mathematical or physical object rotates, which axes remain fixed, which inputs are known, and which result is expected. This guide supports planning rather than providing a completed Python solution.

1. Four questions before coding

1. **What operation is involved?** Does the vector rotate, or do the reference axes rotate?
2. **What are the inputs?** Vector (v_x, v_y), angle and units.
3. **What are the outputs?** Rotated components, components in another frame, norm, or comparison.
4. **How will it be checked?** Manual substitution, norm preservation or inverse transformation.

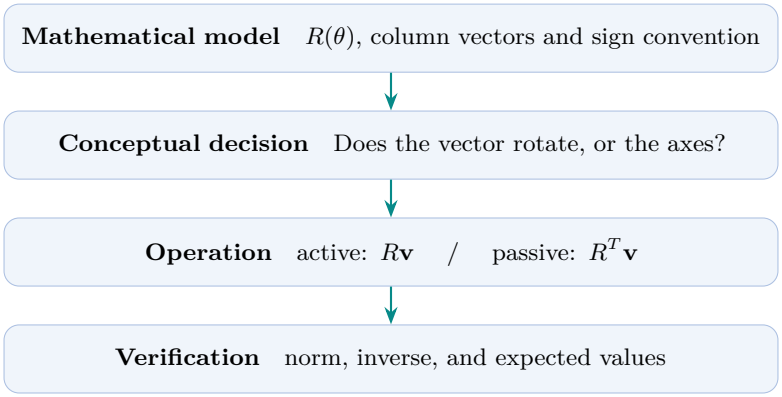
2. Conceptual pseudocode for an active rotation

```
START
1.  IDENTIFY vector and angle (+ counterclockwise).
2.  CONVERT degrees to radians.
3.  COMPUTE cosine and sine.
4.  BUILD the rotation matrix.
5.  MULTIPLY matrix by vector.
6.  COMPARE initial and transformed norms.
7.  DISPLAY vector and verification.
END
```

3. Pseudocode for a passive change of basis

```
START
1.  IDENTIFY vector in global axes.
2.  IDENTIFY rotation of axes (not of vector).
3.  BUILD the same matrix R.
4.  USE the transpose R.T for local components.
5.  RECOVER global components using R.
6.  CHECK components and vector norm.
END
```

4. Reasoning sequence in a flow diagram



5. How to organise the EX29 sections

Section	Suggested pseudocode planning
A0–A2	Build R, multiply, check manually and invert.
A3	Compose two matrices; compare with a single rotation.
B	Include physical units and check force magnitude.
C	Keep the vector fixed, apply transpose, recover global components.

6. Programming tips

- Use names that clarify whether a quantity is global, local, original or rotated.
- Separate operations during learning; do not compress everything into one expression.
- Avoid direct `float == float` checks when approximate trigonometric values are expected.
- Use `shape` when debugging array dimensions.
- Advanced functions, JSON and automated tests are not required in this class.